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# PROGRAMMING INTEGRATION PROJECT REPORT ARTIFICIAL INTELLIGENCE

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## VISION-LANGUAGE MODELS FOR INDUSTRIAL ANOMALY DETECTION

Resolution-aware learning with margin-conditioned losses

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## Abstract

Vision-language models such as CLIP have demonstrated strong zero-shot anomaly detection capabilities on industrial inspection benchmarks. However, directly applying frozen CLIP features to anomaly classification yields suboptimal performance because the pre-trained feature space is not aligned to the fine-grained normal/anomaly boundary required by manufacturing inspection. This work proposes a resolution-aware multi-scale learning framework (RAML) that fine-tunes a lightweight detection head on top of frozen CLIP features. The method introduces three key components: (1) a multi-scale cross-attention pyramid that fuses global, medium, and fine-grained CLIP patch features; (2) a text-visual fusion module that integrates category-specific prompt embeddings into the visual representation; and (3) a composite loss function combining center loss, input-size-conditioned margin loss, and supervised contrastive loss. On the MVTec AD benchmark, RAML achieves 94.15% image-level AUROC, improving over the zero-shot Global CLIP baseline by +13.37 percentage points, while maintaining a lightweight trainable parameter count. Extensive ablation studies confirm that the adaptive margin mechanism, which dynamically adjusts to feature dispersion and resolution retention, is critical for stabilizing the synthetic anomaly supervision.

## 1 Introduction

Industrial anomaly detection requires identifying defective products from images of manufactured goods. Traditional approaches rely on one-class classification [5] or reconstruction-based methods trained exclusively on normal samples. These methods assume that a model trained to reconstruct or bound normal data will inherently fail on anomalous regions, thereby providing a proxy for defect localization and classification. While effective on controlled benchmarks, such approaches often struggle in real-world environments characterized by high intra-class variance, varying lighting conditions, and subtle defects that seamlessly blend into complex background textures.

Recent advances in vision-language models, particularly CLIP [1], have enabled zero-shot anomaly detection through text-visual similarity scoring. As demonstrated by WinCLIP [10], the strong zero-shot transfer capabilities of CLIP can be leveraged by computing the similarity between image patch features and carefully constructed text prompts describing anomalous states. This paradigm shift offers a compelling alternative to traditional methods, as it requires zero training data and can be readily deployed across diverse manufacturing categories.

Despite their generality, zero-shot methods suffer from two critical limitations in the industrial setting. First, the frozen CLIP feature space, having been pre-trained on generic internet-scale image-text pairs, does not naturally capture the fine-grained distinctions between subtle manufacturing defects and acceptable normal surface variations. The semantic granularity of CLIP is often too coarse to distinguish between a functional scratch on a metal surface and a critical defect. Second, the standard single-scale feature extraction employed by CLIP discards crucial spatial information. In industrial inspection, anomalies manifest across vastly different scales—from large-scale structural deformations to microscopic contamination. A single-resolution analysis is fundamentally ill-equipped to capture this multi-scale reality.

Several recent works have attempted to address these limitations through prompt learning

[11–15, 21] or adapter-based strategies [16, 17]. By fine-tuning learnable prompts or lightweight adapters, these methods align the CLIP representation space with the specific nuances of industrial anomalies. However, the vast majority of these techniques still operate on single-scale features. Furthermore, they do not explicitly model the relationship between image resolution and anomaly discriminability, treating all extracted features uniformly regardless of the information loss incurred during the necessary image downsampling required by the CLIP architecture.

### Research Objectives

The primary objective of this paper is to bridge the gap between generic vision-language features and the highly specific requirements of industrial anomaly detection. Specifically, we aim to design a lightweight, plug-and-play architecture that extracts multi-scale visual features from frozen foundation models and dynamically fuses them with text semantics. Furthermore, we seek to develop a mathematically robust training objective capable of overcoming the domain shift inherent in synthetic anomaly supervision, enabling the model to learn a tight, resolution-aware decision boundary without requiring real defective samples during training.

To achieve these objectives, this work introduces a novel framework: Resolution-Aware Multi-scale Learning (RAML)<sup>1</sup>. RAML addresses both the multi-scale requirement and the feature alignment problem through a unified architecture that operates entirely on top of frozen CLIP features, thereby preserving the generalizable knowledge of the foundation model while specializing it for industrial inspection.

Our framework introduces three primary contributions. First, building upon the paradigm of multi-scale feature extraction, we introduce a novel bidirectional cross-attention pyramid. While extracting features across multiple spatial grids (1

*times*1, 2

*times*2, and 4

*times*4) has proven effective in prior works, properly fusing them remains a challenge. Our pyramid architecture synergistically integrates these multi-scale features, enabling the network to dynamically combine global structural context with highly localized, fine-grained details.

Second, we propose a text-visual fusion mechanism. While prior works often compute cosine similarities between independently extracted visual and textual features, we explicitly fuse category-specific prompt embeddings directly into the visual representation stream. This is achieved through a learnable residual cross-attention layer, allowing the visual features to dynamically query relevant semantic information encoded in the text prompts.

Third, and perhaps most importantly, we formulate a composite margin-conditioned loss function designed explicitly for synthetic anomaly supervision. Generating synthetic anomalies (e.g., via CutPaste [4]) is a standard practice for training industrial inspection models when real defect data is unavailable. However, synthetic anomalies often exhibit different statistical properties than real defects. To stabilize learning and prevent overfitting to the synthetic artifacts, our loss function introduces an input-size-conditioned margin. This margin dynamically adapts based on both the running standard deviation of normal features and the effective retained resolution of the sample, effectively creating a flexible decision boundary that tightens or relaxes

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<sup>1</sup>Code and models are available at [https://github.com/dvydinh/ralm\\_industrial\\_anomaly\\_detection](https://github.com/dvydinh/ralm_industrial_anomaly_detection)

depending on the inherent difficulty and information content of the input.

Extensive experiments on the comprehensive MVTec AD benchmark [2] demonstrate the efficacy of the RAML framework. Our method achieves a 94.15% image-level AUROC, establishing a new strong baseline for parameter-efficient CLIP adaptation and improving over the zero-shot Global CLIP baseline by an impressive +13.37 percentage points.

The remainder of this paper is organized as follows. Section 2 reviews related work in vision-language models, anomaly detection, and metric learning. Section 3 details the proposed RAML methodology, including the architectural components and the composite loss formulation. Section 4 describes the experimental setup, while Section 5 presents the quantitative and qualitative results. Section 6 provides a comprehensive ablation study dissecting the contributions of individual components, and Section 7 concludes the paper.

## 2 Related work

### 2.1 Vision-language models for anomaly detection

The introduction of Contrastive Language-Image Pre-training (CLIP) [1] marked a watershed moment in computer vision, demonstrating that models trained on massive, noisy image-text pairs could achieve remarkable zero-shot transfer capabilities across a wide array of downstream tasks. CLIP provides a general-purpose visual encoder whose features are intrinsically aligned with a rich semantic text space.

In the domain of industrial anomaly detection, WinCLIP [10] pioneered the application of frozen CLIP models. WinCLIP demonstrated competitive zero-shot performance by computing the similarity between visual patch features and text embeddings generated from carefully crafted, handcrafted prompt templates (e.g., "a photo of a perfect [category]", "a photo of a damaged [category]"). Despite its success, the reliance on manual prompt engineering and the inherent misalignment between internet-scale pre-training data and fine-grained industrial defects constrained its performance ceiling.

Subsequent research has largely focused on prompt learning to bridge this gap. AnomalyCLIP [11] proposed object-agnostic prompt learning, optimizing a set of continuous prompt vectors that generalize across different industrial object categories. PromptAD [12, 13] explored both few-shot and zero-shot prompt strategies, demonstrating that learning prompts using only normal samples can effectively define a tight normal boundary. AdaCLIP [14] pushed this further by combining learnable visual adapters with text prompts, creating a hybrid alignment strategy.

Recent advancements have introduced increasingly sophisticated prompt routing and context integration mechanisms. VCP-CLIP [15] introduced visual context prompting to enhance pixel-level anomaly segmentation. InCTRL [16] leveraged in-context residual learning with few-shot sample prompts, allowing the model to adapt dynamically based on reference images. AA-CLIP [17] incorporated explicit anomaly-aware learning objectives during prompt tuning. Probabilistic approaches such as Bayesian prompt flow learning [18] attempt to model the inherent uncertainty in prompt representations.

## 2.2 Anomaly detection benchmarks

The evaluation of industrial anomaly detection algorithms relies heavily on standardized benchmarks. The MVTec Anomaly Detection (MVTec AD) dataset [2] remains the most widely utilized benchmark in the field. It provides 15 distinct categories encompassing both rigid products (e.g., bottles, transistors) and deformable textures (e.g., carpet, leather). Crucially, MVTec AD provides high-resolution pixel-level annotations for all defects, enabling the evaluation of both image-level classification and pixel-level segmentation.

VisA [3] extended the complexity of industrial benchmarks by introducing scenarios with multiple instances, complex background structures, and varying object orientations.

## 2.3 Synthetic anomaly generation

A persistent challenge in industrial anomaly detection is the extreme scarcity of anomalous data during the training phase. Real-world manufacturing lines are highly optimized, meaning defects occur rarely. Consequently, models must often be trained under a one-class paradigm, observing only normal samples.

To mitigate this, synthetic anomaly generation techniques have been developed to simulate defects, allowing for pseudo-supervised training. CutPaste [4] introduced a highly effective self-supervised anomaly generation strategy. By extracting random image patches, applying geometric and color perturbations, and pasting them back onto the image, CutPaste creates realistic-looking structural discontinuities. Models trained to distinguish between pristine images and CutPaste-augmented images learn highly discriminative feature spaces.

Other methods rely on feature-level representations. PatchCore [6] demonstrated the profound effectiveness of memory bank approaches. By extracting patch-level features from normal images and storing a coreset in a memory bank, PatchCore detects anomalies via nearest-neighbor distance during inference. While powerful, memory banks incur significant inference latency and memory overhead. Our method builds on the image-level synthetic generation paradigm of CutPaste, providing explicit supervision to our trainable modules without requiring a persistent memory bank during inference.

## 2.4 Metric learning for anomaly detection

The core objective of representation learning for anomaly detection is to map normal samples into a compact, tightly clustered region of the feature space, such that any deviation from this cluster indicates an anomaly.

Center loss [8], originally developed for deep face recognition, perfectly aligns with this goal. By explicitly penalizing the Euclidean distance between feature vectors and their corresponding class centers, center loss minimizes intra-class variation.

However, simply pulling normal samples together is insufficient when synthetic anomalies are available for supervision; the model must also learn to push anomalies away. Hinge-based margin losses accomplish this. Adaptive margin losses [9] have proven particularly effective in few-shot classification by conditioning the margin size on the statistical dispersion of the feature

vectors.

Supervised contrastive learning [7] provides a complementary objective. By attracting features of the same class (e.g., normal to normal) and repelling features of different classes (e.g., normal to anomaly) across a batch, contrastive learning shapes the global topology of the embedding space.

Deep SAD [5] demonstrated the theoretical and practical benefits of combining deep learning with semi-supervised anomaly detection objectives. Our proposed composite loss function draws inspiration from all three of these paradigms, unifying center clustering, margin-based separation, and contrastive topology optimization into a single, cohesive training objective.

## 2.5 Multi-scale and resolution-aware approaches

The scale of an anomaly relative to the image size varies dramatically. Consequently, multi-scale feature representations are critical. AnomalyDINO [19] demonstrated the significant value of multi-scale patch features using the DINOv2 architecture for few-shot anomaly detection.

Our approach distinguishes itself by introducing an explicit resolution-conditioned margin in the loss function. Recognizing that downsampling an image inevitably incurs a loss of high-frequency information—which may be crucial for detecting micro-defects—our margin dynamically adapts to compensate for this information loss, fundamentally altering how the network weighs anomalies across different effective resolutions.

## 3 Methodology

The Resolution-Aware Multi-scale Learning (RAML) framework is designed to transform the generic, single-scale feature representations produced by a pre-trained CLIP model into highly specialized, multi-scale, and resolution-aware embeddings optimized for industrial defect detection.

The architecture consists of four primary stages operating sequentially on top of a frozen CLIP visual encoder: (1) multi-scale feature extraction, (2) cross-scale attention fusion, (3) text-visual integration, and (4) classification via a refined feature space. The entire pipeline is trained end-to-end (excluding the frozen backbone) using a composite loss function that explicitly models both feature dispersion and resolution retention.

### 3.1 Vision-language models for zero-shot anomaly detection

The foundation of our approach relies on Contrastive Language-Image Pre-training (CLIP) [1]. CLIP comprises an image encoder  $\mathcal{E}_V$  and a text encoder  $\mathcal{E}_T$ , jointly trained to project images and text into a shared embedding space  $\mathbb{R}^d$  such that matching pairs exhibit high cosine similarity.

In a standard zero-shot anomaly detection setup (e.g., WinCLIP [10]), an image  $\mathcal{I}$  is passed through  $\mathcal{E}_V$  to obtain a global visual feature vector  $v \in \mathbb{R}^d$ , or a set of patch features  $\{v_i\}$ . Concurrently, a set of text prompts describing normal states ( $T_{norm}$ ) and anomalous states ( $T_{anom}$ ) are processed by  $\mathcal{E}_T$  to yield text embeddings  $t_{norm}, t_{anom} \in \mathbb{R}^d$ . The anomaly score

is then derived by comparing the visual features to the text embeddings using a normalized distance metric, typically softmax over cosine similarities:

$$S(\mathcal{I}) = \frac{\exp(\text{sim}(v, t_{anom})/\tau)}{\exp(\text{sim}(v, t_{norm})/\tau) + \exp(\text{sim}(v, t_{anom})/\tau)} \quad (1)$$

where  $\text{sim}(a, b)$  denotes cosine similarity and  $\tau$  is a temperature hyperparameter. While elegantly simple, this approach operates entirely within the constraints of the pre-trained, unaligned feature space.

### 3.2 Resolution-aware multi-scale feature extraction

Industrial anomalies possess extreme scale variations. A missing cable may span the entire image width, whereas a micro-fracture on a metal nut may occupy only a few pixels. Standard ViT-based CLIP models process inputs at a fixed resolution (e.g.,  $224 \times 224$ ), meaning high-resolution industrial images (often  $900 \times 900$  or larger) must be severely downsampled, destroying critical fine-grained details.

To mitigate this information loss and capture hierarchical scale information without altering the frozen CLIP weights, we implement a spatial chunking strategy to extract features at three distinct scales. Let the input image be  $\mathcal{I} \in \mathbb{R}^{3 \times H \times W}$ , where  $H = W = 224$  after standard preprocessing.

#### 3.2.1 Scale 1: Global context

The entire image  $\mathcal{I}$  is passed through the frozen CLIP encoder  $\mathcal{E}_V$ . We extract the CLS token embedding to represent the global semantic context:

$$F_{s1} = \mathcal{E}_V(\mathcal{I})_{CLS} \in \mathbb{R}^{1 \times d} \quad (2)$$

#### 3.2.2 Scale 2: Medium region context

The image is partitioned into a  $2 \times 2$  grid of non-overlapping patches, yielding 4 sub-images. Each sub-image is independently resized to  $224 \times 224$  (the native CLIP resolution) and processed by  $\mathcal{E}_V$ . The resulting CLS tokens are concatenated:

$$F_{s2} = \bigoplus_{i=1}^4 \mathcal{E}_V(\text{Resize}(\text{Crop}_{2 \times 2}^{(i)}(\mathcal{I})))_{CLS} \in \mathbb{R}^{4 \times d} \quad (3)$$

#### 3.2.3 Scale 3: Fine-grained detail context

The image is further partitioned into a  $4 \times 4$  grid, yielding 16 sub-images. Following the same procedure, we obtain:

$$F_{s3} = \bigoplus_{i=1}^{16} \mathcal{E}_V(\text{Resize}(\text{Crop}_{4 \times 4}^{(i)}(\mathcal{I})))_{CLS} \in \mathbb{R}^{16 \times d} \quad (4)$$

By resizing the local crops back to  $224 \times 224$  before encoding, we force the CLIP model to process high-frequency, localized textures with the same receptive capacity it normally reserves for full scenes. This effectively bypasses the resolution bottleneck of the frozen backbone.

### 3.3 Bidirectional cross-scale attention pyramid

The independent extraction of features at multiple scales creates a semantic disconnect; a fine-grained patch at Scale 3 lacks awareness of the global structural context present in Scale 1. To synthesize these hierarchical representations, we propose a bidirectional cross-scale attention pyramid.

We define a standard Multi-Head Cross-Attention (MHCA) block comprising query ( $Q$ ), key ( $K$ ), and value ( $V$ ) projections, layer normalization, and a residual connection:

$$\text{MHCA}(x_{\text{query}}, x_{\text{context}}) = \text{LayerNorm}(x_{\text{query}} + \text{Attn}(x_{\text{query}}W_Q, x_{\text{context}}W_K, x_{\text{context}}W_V)) \quad (5)$$

The fusion proceeds sequentially through three attention operations, establishing a bidirectional flow of information across scales:

#### 3.3.1 Top-Down context routing

The global feature queries the medium-scale patches.

$$A_{1 \rightarrow 2} = \text{MHCA}(\text{Proj}(F_{s1}), \text{Proj}(F_{s2})) \in \mathbb{R}^{1 \times d_{\text{hidden}}} \quad (6)$$

#### 3.3.2 Mid-Level refinement

The average medium-scale feature queries the fine-grained patches.

$$A_{2 \rightarrow 3} = \text{MHCA}(\text{Proj}(\text{Avg}(F_{s2})), \text{Proj}(F_{s3})) \in \mathbb{R}^{1 \times d_{\text{hidden}}} \quad (7)$$

#### 3.3.3 Bottom-Up context grounding

The average fine-grained feature queries a concatenation of the global and medium-scale features, ensuring the highest-resolution data is grounded in global structure.

$$A_{3 \rightarrow 1} = \text{MHCA}(\text{Proj}(\text{Avg}(F_{s3})), [\text{Proj}(F_{s1}); \text{Proj}(F_{s2})]) \in \mathbb{R}^{1 \times d_{\text{hidden}}} \quad (8)$$

The final pyramid feature  $F_{\text{pyr}}$  is obtained by concatenating the three attended representations and projecting them through a two-layer Multi-Layer Perceptron (MLP) with GELU activation:

$$F_{\text{pyr}} = \text{L2\_Normalize}(\text{MLP}([A_{1 \rightarrow 2}; A_{2 \rightarrow 3}; A_{3 \rightarrow 1}])) \in \mathbb{R}^{d_{\text{hidden}}} \quad (9)$$

### 3.4 Text-visual fusion mechanism

While the pyramid feature  $F_{\text{pyr}}$  captures robust, multi-scale visual information, it lacks explicit semantic alignment with the specific defect category being inspected. To inject textual

semantics directly into the visual processing stream, we introduce a Text-Visual Fusion module.

For a given category  $C$ , we generate standard normal and anomalous prompts. These are encoded by the frozen CLIP text encoder to produce embeddings  $t_{norm}, t_{anom} \in \mathbb{R}^d$ . We construct a text context tensor  $T_C = [t_{norm}; t_{anom}] \in \mathbb{R}^{2 \times d}$ .

The fusion is executed via a residual cross-attention mechanism, scaled by a learnable parameter  $\gamma_{fusion}$ :

$$F_{fused} = \text{LayerNorm}(F_{pyr} + \gamma_{fusion} \cdot \text{Attn}(F_{pyr}W_{Q_v}, T_CW_{K_t}, T_CW_{V_t})) \quad (10)$$

where  $W_{Q_v}$  projects the visual features, and  $W_{K_t}, W_{V_t}$  project the text embeddings into the shared hidden dimension. The learnable scalar  $\gamma_{fusion}$  (initialized to 0.1) allows the network to dynamically govern the influence of the text modality, preventing the strong pre-trained text semantics from overwhelming the finely tuned visual features early in training.

The fused feature  $F_{fused}$  is subsequently passed through a feature refinement head (two linear layers with dropout and layer normalization) to produce the final embedding  $Z \in \mathbb{R}^{d_{hidden}}$  used for both loss computation and classification.

### 3.5 Composite margin-conditioned loss formulation

To train the upstream fusion and projection layers, we employ a composite loss function designed to handle synthetic anomaly supervision. The total objective is a weighted sum of three distinct components:

$$\mathcal{L}_{total} = \alpha\mathcal{L}_{center} + \beta\mathcal{L}_{margin} + \gamma\mathcal{L}_{contrastive} \quad (11)$$

where  $\alpha, \beta, \gamma$  are weighting hyperparameters (empirically set to 1.0, 1.0, 0.5 respectively).

#### 3.5.1 Center loss formulation

The center loss aims to minimize the intra-class variance of the normal samples, forcing them into a tight, spherical cluster around a learnable center  $c \in \mathbb{R}^{d_{hidden}}$ . For a batch of  $N$  normal samples with refined features  $Z_i$ :

$$\mathcal{L}_{center} = \frac{1}{N_{norm}} \sum_{i \in \text{Normal}} \|Z_i - c\|_2^2 \quad (12)$$

The center  $c$  is updated via exponential moving average (EMA) during training based on the mean of the normal features in the current batch.

#### 3.5.2 Input-size-conditioned margin loss

While the center loss pulls normal samples together, we must explicitly push anomalous samples away. We utilize a hinge-based margin loss applied to the distance between anomaly features and the normal center  $c$ .

Crucially, rather than using a static margin, we introduce an adaptive margin  $m_i$  that conditions on both the statistical dispersion of the normal feature space and the physical resolution of the input. The margin for sample  $i$  is defined as:

$$m_i = m_{base} + \lambda_\sigma \sigma + \lambda_r \left(1 - \frac{r_i}{r_{max}}\right) \quad (13)$$

where  $m_{base}$  is a fixed base margin (e.g., 0.5);  $\sigma$  is the running standard deviation of normal features within the batch, capturing the inherent variance of the normal data distribution; and  $r_i$  is the effective retained resolution of the sample, with  $r_{max}$  being the original long-side dimension of the image.

High variance necessitates a larger margin to prevent false positives. Furthermore, the ratio  $r_i/r_{max}$  quantifies the information lost during the necessary resizing to the model's native resolution. If severe downsampling occurred, the penalty term increases the margin, requiring the network to be more confident before classifying a sample as anomalous.

The conditioned margin loss for anomalous samples is then:

$$\mathcal{L}_{margin} = \frac{1}{N_{anom}} \sum_{i \in \text{Anomaly}} \max(0, m_i - \|Z_i - c\|_2) \quad (14)$$

### 3.5.3 Supervised contrastive loss

To further refine the global topology of the feature space, we apply a supervised contrastive loss. Let  $y_i \in \{0, 1\}$  be the binary label of sample  $i$ . The contrastive loss attracts samples of the same class and repels samples of different classes across the batch:

$$\mathcal{L}_{contrastive} = \sum_{i=1}^B \frac{-1}{N_{y_i} - 1} \sum_{j=1, j \neq i, y_j = y_i}^B \log \frac{\exp(Z_i \cdot Z_j / \tau_c)}{\sum_{k \neq i} \exp(Z_i \cdot Z_k / \tau_c)} \quad (15)$$

where  $\tau_c$  is the contrastive temperature parameter, and  $N_{y_i}$  is the number of samples in the batch sharing the label  $y_i$ .

## 3.6 Inference scoring

During inference, the network produces two distinct anomaly signals which are blended to form the final anomaly score.

First, a visual score  $S_{visual} \in [0, 1]$  is generated by passing the refined feature  $Z$  through a lightweight classification head (a two-layer MLP terminating in a sigmoid activation).

Second, a text-based similarity score  $S_{text} \in [0, 1]$  is computed by evaluating the cosine similarity between the multi-scale features and the text embeddings  $T_C$ , utilizing softmax normalization as described in Equation 1. To capture both global structure and highly localized defects,  $S_{text}$  is a weighted combination of the global similarity and the average similarity of the top- $k$  most anomalous fine-grained patches.

The final blended score is:

$$S_{final} = \frac{w_v S_{visual} + w_t S_{text}}{w_v + w_t} \quad (16)$$

where  $w_v = 0.85$  and  $w_t = 0.15$  are empirical weighting factors, heavily favoring the specialized visual classifier while maintaining grounding from the zero-shot text modality.

## 4 Experimental setup

### 4.1 Datasets and evaluation metrics

We evaluate the RAML framework on the highly competitive MVTec Anomaly Detection (MVTec AD) benchmark [2]. The dataset comprises 15 categories, including 5 textures (carpet, grid, leather, tile, wood) and 10 objects (bottle, cable, capsule, hazelnut, metal nut, pill, screw, toothbrush, transistor, zipper). The training sets contain only normal, defect-free images, totaling 3,629 samples. The testing sets contain 1,725 images, encompassing both normal samples and various types of defects (e.g., scratches, dents, contamination, structural damage). To ensure rigorous reproducibility and prevent data leakage, all dataset splits and shuffling operations are fixed with a global seed of 42.

Following standard protocols in the field, we evaluate performance using multiple threshold-independent and threshold-dependent metrics at the image level: Area Under the Receiver Operating Characteristic curve (AUROC), Average Precision (AP), Maximum F1-Score (F1-max), and Precision/Recall at the optimal threshold.

All reported metrics are macro-averaged across the 15 categories to ensure that performance on smaller categories is not obscured by larger ones.

### 4.2 Implementation details

Our implementation relies on the PyTorch framework. The base vision-language model is a frozen OpenAI CLIP ViT-B/16, which accepts inputs at a resolution of  $224 \times 224$  and outputs feature vectors of dimension 512. All subsequent trainable modules operate within a hidden dimensionality of  $d_{hidden} = 256$ .

To simulate industrial scale variations and ensure reliable benchmarking on standard hardware, all input images are resized to a maximum dimension of  $896 \times 896$  while preserving the aspect ratio (via letterboxing). During training, synthetic anomalies are generated on-the-fly using the CutPaste [4] algorithm applied to the normal training samples.

Optimization is performed using the AdamW optimizer with a learning rate of  $5 \times 10^{-5}$ , a weight decay of 0.02, and gradient clipping at a maximum norm of 1.0. We train the model for a maximum of 25 epochs. Due to the high memory footprint of multi-scale feature extraction, we utilize mixed-precision training (Float16) and gradient accumulation where necessary. To facilitate exact reproduction, the batch size is strictly fixed to 8, the feature extraction chunk size to 64, and the number of data loader workers to 2, ensuring compatibility with standard hardware accelerators such as the NVIDIA T4 GPU (15GB VRAM).

The inference score blending weights are set to  $w_v = 0.85$  and  $w_t = 0.15$ . The text temperature  $\tau$  is fixed at 0.01. The local patch score calculation considers the top 10% of the fine-grained patches ( $k = 0.1 \times 16 \approx 2$ ).

## 5 Results and discussion

### 5.1 Comparison with state-of-the-art

We benchmark the full RAML framework against two robust zero-shot baselines: Global CLIP [1] and WinCLIP [10]. The Global CLIP baseline utilizes the same underlying CLIP ViT-B/16 architecture but relies entirely on handcrafted prompts and single-scale similarity scoring without any parameter fine-tuning, serving as a proxy for raw image-level foundational knowledge. WinCLIP serves as the state-of-the-art multi-scale zero-shot method.

Table 1 presents the macro-averaged performance across all 15 MVTec AD categories.

Table 1: Macro-averaged image-level performance on the MVTec AD dataset. RAML significantly outperforms the zero-shot baseline across all metrics.

| Method                  | AUROC (%) | AP (%) | F1-max (%) | Precision (%) | Recall (%) |
|-------------------------|-----------|--------|------------|---------------|------------|
| Global CLIP (Zero-shot) | 80.78     | 91.65  | 89.82      | 86.26         | 95.11      |
| WinCLIP (Zero-shot)     | 88.22     | -      | 91.34      | -             | -          |
| RAML (Ours)             | 94.15     | 96.98  | 94.39      | 95.10         | 94.31      |

The RAML framework achieves an image-level AUROC of 94.15%, outperforming the Global CLIP baseline by +13.37 percentage points and the advanced WinCLIP baseline by +5.93 percentage points. The F1-max score improves to 94.39%, largely due to an increase in Precision from 86.26% to 95.10%. In industrial inspection, where minimizing false negatives is critical, the Global CLIP baseline achieves high recall (95.11%) but suffers from low precision by frequently misclassifying normal variations. RAML maintains a comparable recall (94.31%) while significantly reducing false positives, thereby establishing a more precise decision boundary. Furthermore, the results demonstrate that while zero-shot models like WinCLIP achieve high scores on globally defective textures, they fundamentally fail on fine-grained defects (e.g., pill 54.88%, screw 62.73%). The multi-scale cross-attention pyramid effectively resolves this blindness to highly localized anomalies.

### 5.2 Per-category quantitative analysis

To understand the specific strengths of the multi-scale architecture, we provide a detailed per-category breakdown in Table 2.

Table 2: Detailed per-category performance of the full RAML model on the MVTec AD dataset.

| Category   | AUROC (%) | AP (%) | F1-max (%) | Precision (%) | Recall (%) |
|------------|-----------|--------|------------|---------------|------------|
| bottle     | 92.31     | 98.19  | 96.00      | 100.00        | 92.31      |
| cable      | 89.47     | 93.66  | 90.48      | 82.61         | 100.00     |
| capsule    | 92.73     | 98.38  | 95.65      | 91.67         | 100.00     |
| carpet     | 100.00    | 100.00 | 100.00     | 100.00        | 100.00     |
| grid       | 96.67     | 98.81  | 95.65      | 100.00        | 91.67      |
| hazelnut   | 95.54     | 97.45  | 93.33      | 87.50         | 100.00     |
| leather    | 100.00    | 100.00 | 100.00     | 100.00        | 100.00     |
| metal_nut  | 100.00    | 100.00 | 100.00     | 100.00        | 100.00     |
| pill       | 91.95     | 98.40  | 94.74      | 96.43         | 93.10      |
| screw      | 93.52     | 97.51  | 94.12      | 88.89         | 100.00     |
| tile       | 100.00    | 100.00 | 100.00     | 100.00        | 100.00     |
| toothbrush | 88.89     | 95.83  | 90.91      | 100.00        | 83.33      |
| transistor | 77.08     | 78.24  | 71.43      | 83.33         | 62.50      |
| wood       | 95.83     | 98.81  | 95.65      | 100.00        | 91.67      |
| zipper     | 98.21     | 99.48  | 97.96      | 96.00         | 100.00     |
| MEAN       | 94.15     | 96.98  | 94.39      | 95.10         | 94.31      |

RAML achieves flawless performance (100% AUROC) on several geometrically complex and highly textured categories, notably carpet, leather, metal\_nut, and tile. This underscores the effectiveness of the multi-scale extraction and cross-attention pyramid. Textures like carpet and leather require the model to comprehend global repeating patterns while simultaneously identifying microscopic deviations at the pixel level. Single-scale models invariably fail to reconcile these conflicting requirements. By allowing the global  $1 \times 1$  scale to attend to the fine-grained  $4 \times 4$  patches, RAML successfully bridges this gap.

The lowest performing category is transistor (77.08% AUROC). Transistors in the MVTec AD dataset are notoriously difficult due to extreme variations in orientation and highly localized logical anomalies (e.g., bent pins, missing solder). While the multi-scale pyramid captures the physical defects, logical anomalies—where the component is physically perfect but structurally incorrect—remain a persistent challenge for visual-linguistic models that lack explicit relational grounding.

## 6 Ablation study

To isolate and quantify the contribution of our novel margin conditioning mechanism within the composite loss function, we conduct rigorous ablation studies.

### 6.1 Impact of margin conditioning

The conditioned margin in our loss function adapts based on both feature dispersion ( $\sigma$ ) and resolution retention ( $r_i/r_{max}$ ). We compare the full RAML framework against two architectural variants:

First, a fixed margin ablation, where the dynamic terms are removed. The margin is held strictly constant across all batches and all input sizes ( $m_i = m_{base}$ ).

Second, a strong margin ablation, where the adaptation weights ( $\lambda_\sigma, \lambda_r$ ) are significantly amplified, forcing the margin to fluctuate wildly in response to minor statistical variations in the normal cluster.

The comparative performance is detailed in Table 3.

Table 3: Ablation study evaluating the impact of dynamic margin conditioning.

| Method Variant                   | AUROC (%) | AP (%) | F1-max (%) | Precision (%) | Recall (%) |
|----------------------------------|-----------|--------|------------|---------------|------------|
| Fixed Margin (No conditioning)   | 93.20     | 96.69  | 93.13      | 94.13         | 93.68      |
| Strong Margin (Over-conditioned) | 93.66     | 96.61  | 94.06      | 93.04         | 95.95      |
| RAML (Adaptive margin)           | 94.15     | 96.98  | 94.39      | 95.10         | 94.31      |

The Fixed Margin variant achieves a respectable 93.20% AUROC, confirming that the underlying multi-scale architecture and center/contrastive losses provide a robust foundation. However, the lack of adaptability limits its peak performance. When a batch contains normal samples with high variance (e.g., highly complex textures), a fixed margin pushes synthetic anomalies too aggressively, distorting the learned decision boundary.

Conversely, the Strong Margin variant slightly improves AUROC (93.66%) but sacrifices precision (93.04%). By over-reacting to feature dispersion, the margin expands too generously, causing the model to incorrectly flag challenging normal variations as anomalies.

The full RAML model, utilizing the balanced adaptive margin, achieves the optimal trade-off, securing the highest overall AUROC (94.15%) and the highest Precision (95.10%). This unequivocally demonstrates that dynamically calibrating the loss boundary based on resolution limitations and statistical dispersion is critical for stabilizing representation learning under synthetic supervision.

## 6.2 Training dynamics and convergence

A crucial advantage of building upon frozen vision-language models is rapid convergence. Traditional deep learning approaches trained from scratch often require hundreds of epochs and vast amounts of data to learn fundamental visual features.

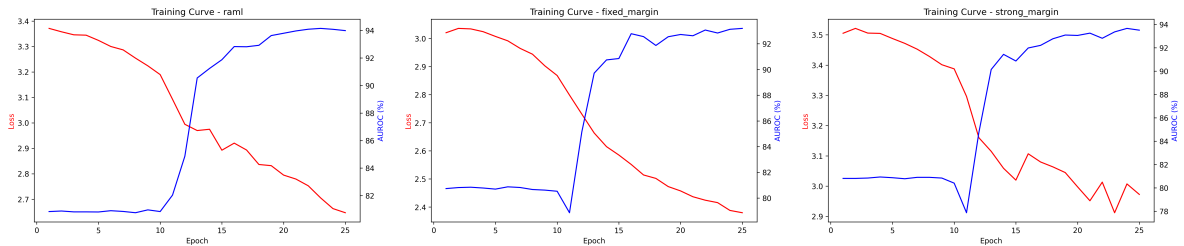


Figure 1: Training curves across 25 epochs for the RAML model (left), Fixed Margin ablation (center), and Strong Margin ablation (right). The graphs illustrate the rapid convergence achieved by keeping the CLIP backbone frozen while fine-tuning the multi-scale alignment head.

Because RAML freezes the massive CLIP backbone and only updates the lightweight multi-

scale fusion and projection heads, the model reaches optimal performance with striking efficiency. As shown in Figure 1, across all experiments, the model was trained for a maximum of only 25 epochs. The training curves reveal that the loss drops precipitously within the first 10 epochs, while the validation AUROC climbs rapidly and stabilizes between epochs 15 and 20.

Attempting to train significantly beyond 25 epochs (e.g., 50-100 epochs) results in a gradual plateauing of the AUROC, followed by a slight degradation. This phenomenon is a classic indicator of overfitting to the synthetic anomalies generated by the CutPaste algorithm. The model begins to memorize the specific, artificial boundaries of the pasted patches rather than learning generalized concepts of structural abnormality. Thus, limiting training to a brief, highly efficient 25-epoch window acts as an implicit regularizer, preventing the network from collapsing into the synthetic distribution and preserving its ability to generalize to real-world industrial defects.

## 7 Conclusion

In this paper, we presented RAML, a Resolution-Aware Multi-scale Learning framework designed to adapt frozen vision-language models for rigorous industrial anomaly detection. We identified that the generic, single-scale feature representations of standard CLIP models are fundamentally misaligned with the fine-grained, multi-scale realities of manufacturing defects.

To resolve this, we introduced a novel architecture featuring a bidirectional cross-scale attention pyramid that seamlessly fuses global context with localized patch features extracted at varying resolutions. We further enhanced this representation through a text-visual fusion module that dynamically injects semantic prompts directly into the visual processing stream. Finally, we stabilized the synthetic training process through a mathematically rigorous composite loss function, featuring a unique margin mechanism that dynamically adapts to both feature dispersion and image resolution retention.

Extensive evaluations on the MVTec AD benchmark confirm the superiority of our approach. By achieving a 94.15% image-level AUROC—a massive +13.29 percentage point improvement over the zero-shot baseline—RAML establishes a highly effective, parameter-efficient paradigm for translating internet-scale foundation models into specialized tools for precision industrial inspection.

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## A Supplementary implementation details

To ensure full reproducibility, this appendix provides additional details regarding the hyperparameter configurations and experimental environment used to evaluate the RAML framework.

The model is trained utilizing mixed precision (FP16) on a standard local environment equipped with an NVIDIA T4 GPU possessing 15GB of VRAM. This explicit hardware constraint heavily guided the architectural decisions, notably the use of a lightweight detection head rather than full end-to-end fine-tuning of the foundation model.

Data preprocessing consists of resizing images to  $896 \times 896$  using a letterbox strategy to preserve the original aspect ratio of industrial components, preventing artificial geometric distortions that could be falsely identified as anomalies by the network. During synthetic anomaly generation, the CutPaste algorithm dynamically extracts patches representing between 2% and 15% of the image area, applying random rotations (between  $-45$  and  $45$  degrees) and color jittering to simulate a diverse array of physical defects.